Longevity. The allostatic load of dietary restriction.

Restriction of food intake by 10-50% of ad libitum on a per unit of weight or energy content basis can extend the lifespan of a wide variety of species and prevent or delay age-related disease. This review first briefly summarizes the data delineating mortality trajectories of various species' populations maintained on restricted diets to provide insight into the effects of nutrient deprivation on distinct components of the aging process. Next, I discuss a number of important studies that have addressed the question whether it is the lack of calories and/or specific nutrients that determines the longevity response to dietary restriction. Finally, I review the evidence for hormesis as a proximate mechanism underpinning the impact of dietary restriction on lifespan. In aggregate, the currently available demographic data suggest that dietary restriction can both slow the age-related progressive accumulation of cellular damage and also enhance the ability of organisms to cope with irreversible injury. Restriction of essential nutrients as well as calories may affect life expectancy, perhaps in a species specific fashion. Hormesis, i.e. an evolutionary conserved stress response routine providing protection against a wide variety of (other) hazards in response to low levels of stress, is very likely to contribute to the beneficial health effects of dietary restriction.